

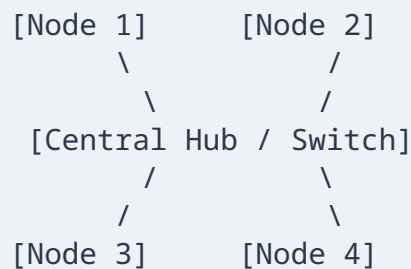
Network Topologies Comprehensive Report

TASK ANALYSIS, ARCHITECTURAL COMPARISON, AND SCENARIO-BASED
DEPLOYMENTS

Task 1: Topologies, Diagrams, and Real-World Applications

Network topologies define how individual devices (nodes) are interconnected within a computer network, dictating data flow, reliability, and scaling dynamics.

1. Star Topology



Real-World App / Service: Enterprise Wi-Fi Networks (e.g., Office Wi-Fi routers connecting laptops and smartphones).

Why: All wireless devices connect directly to a central access point (switch/router), ensuring that if one device disconnects, the rest of the network remains fully operational.

2. Bus Topology



Real-World App / Service: Legacy Ethernet LANs in small older offices / industrial sensor monitoring lines.

Why: Devices share a single central communication backbone (coaxial cable), making it cost-effective for small setups where data traffic is extremely low and sporadic.

3. Mesh Topology



[Node 3] ===== [Node 4]

Real-World App / Service: Cloud Data Center Backbones & Internet Backbone Routing (e.g., AWS multi-region infrastructure).

Why: Every node connects to multiple nodes, providing redundant pathways to guarantee zero downtime and maximum fault tolerance for critical cloud workloads.

Task 2: Comprehensive Comparison Matrix

The table below compares Star, Bus, Ring, and Mesh network topologies across key operational parameters, including pros, cons, and real-world application matches.

Topology	Pros (4 Key Points)	Cons (4 Key Points)	Best Use Case	App Example
Star	• Easy to install & wire • Failure of one cable doesn't affect others • Simple fault isolation at hub • High performance with minimal traffic bottleneck	• Central hub is a single point of failure • Requires more cabling than bus • Higher hardware cost (switches/hubs) • Performance depends heavily on hub capacity	Local Area Networks (LAN), offices, home networks	Instagram (App client connecting to centralized server infrastructure)
Bus	• Low installation cost • Requires very little cable • Easy to understand and set up • Ideal for small temporary networks	• Entire network fails if main cable breaks • Difficult troubleshooting • Performance degrades with heavy traffic • Terminators required at both ends	Small legacy networks, simple educational labs	Zomato (Internal legacy backchannel sensors or simple POS setups)
Ring	• Data travels in one direction (no collisions) • Equal access for all connected devices • Performs better than bus under heavy loads • No central server required to control traffic	• Single device failure breaks the entire ring • Adding/removing nodes disrupts network • Troubleshooting is complex • High latency for distant nodes	Fiber optic backbone rings (SONET), campus loops	Paytm (Legacy token-ring financial transaction loops in regional hubs)
Mesh	• Maximum fault tolerance & redundancy	• Extremely high cabling and setup cost	Mission-critical systems, WANs,	WhatsApp (P2P mesh relay nodes for encrypted

Topology	Pros (4 Key Points)	Cons (4 Key Points)	Best Use Case	App Example
	• No single point of failure • High data security & privacy • Handles massive data traffic effortlessly	• Complex installation and configuration • Requires extensive maintenance • Redundant links increase hardware ports	cloud data centers	routing and media delivery)

Task 3: Multiplayer Mobile Game Server Architecture (PUBG / Free Fire)

When designing a real-time multiplayer battle royale game like **PUBG** or **Free Fire**, selecting the appropriate server network topology is paramount to player retention and competitive integrity.

Recommended Topology: Distributed Star / Hybrid Hub-and-Spoke Regional Server Cluster

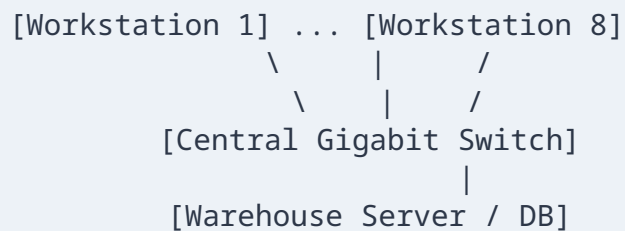
Justification:

- **Speed (Latency):** Players connect directly to regional cloud edge servers in a star configuration, keeping round-trip times (RTT) under 40ms for real-time hit registration.
- **Reliability:** If one regional game instance fails, only that specific lobby is impacted while load balancers instantly redirect traffic, ensuring overall high availability.
- **Cost:** Centralized regional clustering optimizes cloud computing expenditure compared to an expensive full mesh topology while scaling dynamically with concurrent player peaks.

Task 4: Flipkart-Style Warehouse Network Deployment

Scenario: A small e-commerce fulfillment warehouse needs to network 8 computers dedicated to real-time order processing, barcode scanning, and inventory updates.

Recommended Topology: Star Topology



Reasoning:

- **Cost-Effective:** Uses standard Ethernet cables and an affordable 8-port Gigabit switch without excessive hardware overhead.
- **Ease of Troubleshooting:** If a single packing station's cable fails, only that computer goes offline; diagnostics are straightforward at the switch port.
- **Future Expansion:** Additional inventory terminals can be easily plugged into open ports on the switch as warehouse operations scale up.